ORBIFOLD REGULARITY OF KÄHLER–EINSTEIN METRICS ON LOG CANONICAL VARIETIES

by

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Abstract. — Let X be a normal projective log canonical variety with ample canonical class, and let ω_{KE} be the singular Kähler–Einstein metric on it. We prove that ω_{KE} is an orbifold Kähler metric on the orbifold locus of X.

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1. Introduction

The existence of Kähler–Einstein metrics with negative Ricci curvature on smooth canonically polarized varieties was established by Aubin [Aub78] and Yau [Yau78]. In the singular setting, Eyssidieux–Guedj–Zeriahi constructed Kähler–Einstein currents on canonically polarized varieties with klt singularities [EGZ09], and Berman–Guenancia treated log canonical and, more generally, stable varieties [BG14]. Their solution is a full-mass positive current which is a genuine smooth Kähler–Einstein metric on the regular locus.

Let X be a canonically polarized normal projective variety of dimension n with log canonical singularities. Let $\omega_X \in c_1(K_X)$ be a Kähler form on X, and let $d\mu_X$ be an adapted measure associated with a smooth Hermitian metric on the $\mathbb{Q}$-line bundle K_X. We write the Kähler-Einstein metric as $\omega_{\mathrm{KE}} = \omega_X + dd^c\varphi_{\mathrm{KE}}$ on X_{reg}. It is the unique solution to the Monge–Ampère equation

$$\langle(\omega_X + dd^c\varphi_{\mathrm{KE}})^n\rangle = e^{\varphi_{\mathrm{KE}}}d\mu_X, \quad \int_{X_{reg}} \omega_{\mathrm{KE}}^n = K_X^n, \tag{1.1}$$

here $\langle\cdot\rangle$ denotes the non-pluripolar product defined in [BEGZ10].

In general, the behavior of ω_{KE} near singularities is quite mysterious, except for the cases of certain complex hyperbolic cusp singularities [FHJ24] and 2-dimensional Hilbert modular cusp singularities [Kob85]. An easier problem is to consider the behavior of ω_{KE} near orbifold singularities. Denote by

2020 ***Mathematics Subject Classification.*** — 32Q20, 32W20, 14B05, 14J17.
Key words and phrases. — Kähler–Einstein metric, log canonical singularity, orbifold metric, Artin approximation, complex Monge–Ampère equation.

X_{orb} the locus of orbifold singularities. We hope that ω_{KE} is an *orbifold Kähler metric* near $x \in X_{\mathrm{orb}}$, i.e. there exists a local uniformizing chart

$$q : V \longrightarrow U \ni x$$

such that the pullback of $\omega_{\mathrm{KE}}|_{U \cap X_{\mathrm{reg}}}$ extends to a smooth Kähler form on V. The purpose of this paper is to prove the following statement.

Theorem A. — *Let X be a normal projective log canonical variety with K_X ample. Then its negative Kähler–Einstein current ω_{KE} is an orbifold Kähler metric on X_{orb}.*

Indeed, the orbifold modification used in [LT19] together with the Laplacian estimate in [BG14] will give a proof of this theorem. However, the construction of such a modification relies on deep compactification results for stacks. We are going to follow the much more elementary approach used in [GP25], which only involves Artin's approximation theorem.

1.1. Strategy of Proof. — Fix $x \in X_{\mathrm{orb}}$. The strategy is as follows: We first use Artin's theorem to get a local uniformizing chart V at x, then we use Zariski's Main Theorem to compactify it, and get a finite map $f : \overline{V} \to X$. Then take a resolution of $\overline{V}$ to get a smooth variety W and a proper generically finite map

$$\pi : W \longrightarrow X.$$

Since $V \subset \overline{V}_{\mathrm{reg}}$, we see that $\pi|_V$ gives a local uniformizing chart at x, so it suffices to show that $\pi^*\omega_{\mathrm{KE}}$ is a Kähler metric on V. The idea is to consider the Monge-Ampère equation (1.1) on W and apply some Laplacian estimate.

If we write

$$K_W = \pi^* K_X + \sum_i a_i E_i,$$

then the pull-back of the adapted measure $d\mu_X$ under π will have a pole of order $-2a_i$ along E_i. The morphism π factors through a resolution of $\overline{V}$, which as a compactification, need not to be log canonical and may be very singular in general. So a priori we do not have a control on the coefficients a_i. However, using the L^∞ estimate of φ_{KE} obtained in [DGG23], we can show that $a_i \geq -1$, $\forall i \in I$, i.e. $(W, -\sum_i a_i E_i)$ is a sub-lc pair. Then we are able to apply the Laplacian estimate for sub-lc pairs established in [BG14] to conclude.

Acknowledgements. — The author is grateful to Henri Guenancia for helpful discussions on singular Kähler–Einstein metrics and orbifold regularity.

2. Proof of Main Theorem

2.1. Artin approximation. — We use the following Artin's approximation theorem [Art69, Corollary 2.6]:

***Theorem 2.1* (Artin).** — *Let X_1 and X_2 be schemes of finite type over $\mathbb{C}$, and let $x_i \in X_i$. If their completed local rings are isomorphic as complete local $\mathbb{C}$-algebras, then (X_1, x_1) and (X_2, x_2) have a common pointed étale neighborhood. Thus there are a pointed finite-type scheme (U, u) and étale morphisms*

$$(X_1, x_1) \xleftarrow{i_1} (U, u) \xrightarrow{i_2} (X_2, x_2).$$

Now let $x \in X_{\mathrm{orb}}$, then we have a finite and quasi-étale Galois covering $q : \mathbb{C}^n \longrightarrow \mathbb{C}^n/G$, and an isomorphism of germs:

$$(X, x) \simeq (\mathbb{C}^n/G, [0]). \tag{2.1}$$

Let $Y = \mathbb{C}^n/G$ and $y = q(0)$, we therefore have an isomorphism of completed local rings $\hat{\mathcal{O}}_{X,x} \simeq \hat{\mathcal{O}}_{Y,y}$. Then Artin's approximation gives étale maps

$$(X, x) \xleftarrow{i} (U, u) \xrightarrow{j} (Y, y). \tag{2.2}$$

Since quotient singularities are klt, after shrinking U if necessary, we may assume that U is an affine open neighborhood of x, and $i(U) \cap \mathrm{Nklt}(X) = \emptyset$.

Let $V := U \times_Y \mathbb{C}^n$ be the fiber product

$$\begin{array}{ccc} V & \xrightarrow{r} & \mathbb{C}^n \\ \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle q} \\ U & \xrightarrow[j]{} & Y \end{array}$$

Let $p : V \to U$ and $r : V \to \mathbb{A}^n$ be the two projections. The projection p is finite, surjective and quasi-étale since it is the base change of q along the étale map j. The projection r is étale because it is the base change of j. Consequently we see that V is affine and smooth.

Now we consider the morphism

$$F := i \circ p : V \longrightarrow U \longrightarrow X.$$

It is separated and quasi-finite since it is the composition of a finite morphism and an étale morphism. It is also dominant since p is surjective and the étale morphism i is open. Now Zariski's Main Theorem gives an open immersion of V into a scheme $\overline{V}_0$ which is finite over X. Let $\overline{V}$ be the normalization of the closure of V in $\overline{V}_0$, we obtain a factorization

$$V \overset{\iota}{\hookrightarrow} \overline{V} \xrightarrow{f} X, \tag{2.3}$$

of F, where ι is an open immersion, $\overline{V}$ is normal projective and f is finite surjective. In summary, we obtain the following diagram:

$$\begin{array}{ccccc} \overline{V} & & & & \\ \uparrow{\scriptstyle \iota} & \searrow^{f} & & & \\ V & \xrightarrow[p]{} & U & \xrightarrow[i]{} & X \end{array}$$

Since V is an affine open subscheme of $\overline{V}$, it follows from [GW20, Exercise 12.18] that $\overline{V} \setminus V$ has codimension one in $\overline{V}$. Let $B = (\overline{V} \setminus V)_{\mathrm{red}}$ be this boundary divisor. Then the finite map f can ramify along B in general, and the boundary may introduce bad singularities. So $\overline{V}$ may not be log canonical.

Choose a functorial resolution $g : W \longrightarrow \overline{V}$ which is an isomorphism over $\overline{V}_{\mathrm{reg}} \supset V$, put $\pi := f \circ g : W \longrightarrow X$. We write

$$K_W = \pi^* K_X + \sum_{i \in I} a_i E_i, \tag{2.4}$$

where $E = \sum_{i \in I} a_i E_i$ is a $\mathbb{Q}$-divisor supported on the codimension 1 component of the non-étale locus of π, i.e. E_i is either a g-exceptional divisor or the strict transform by g of a component of B. We allow a_i to be 0, so that E_i runs over all exceptional divisors of g and all components of B. In particular, we have $\bigcup_{i \in I} \mathrm{Supp}(E_i) = g^{-1}(\overline{V} \setminus V)$. Since we assumed $i(U) \cap \mathrm{Nklt}(X) = \emptyset$ and V maps onto U, it follows that

$$\pi^{-1}(\mathrm{Nklt}(X)) = g^{-1} \circ f^{-1}(\mathrm{Nklt}(X)) \subset g^{-1}(\overline{V} \setminus V) = \bigcup_{i \in I} \mathrm{Supp}(E_i). \tag{2.5}$$

Obviously $\bigcup_{i \in I} \mathrm{Supp}(E_i)$ is not simple simple crossings now, but one can consider a log resolution $g' : W' \to W$ of the pair $(W, \bigcup_{i \in I} \mathrm{Supp}(E_i))$, and replace $\pi : W \to X$ by $\pi \circ g' : W' \to X$. The equation (2.4) and (2.5) remains true. So without loss of generality, we may assume that $\bigcup_{i \in I} \mathrm{Supp}(E_i)$ is simple normal crossings.

Moreover, by Hironaka's theorem, we can assume g to be a sequence of finitely many blow-ups along smooth centers. Now $f^* K_X$ is an ample $\mathbb{Q}$-line bundle on $\overline{V}$ since it is the pull back under a finite map f of the ample $\mathbb{Q}$-line bundle K_X. Then it is well known that there exists an effective g-exceptional $\mathbb{R}$-divisor $F = \sum c_\alpha F_\alpha$ such that $\pi^* K_X - F$ is ample.

2.2. Control of the discrepancy. — We prove the following proposition:

Proposition 2.2. — *The pair $(W, -E)$ is sub-lc, i.e. the coefficient a_i in (2.4) satisfies $a_i \geq -1$ for all $i \in I$.*

Before proving this proposition, we first recall a lower bound of the Kähler-Einstein potential obtained in [DGG23, Tan26]:

Let $\mathcal{I} \subset \mathcal{O}_X$ be the ideal sheaf of $\mathrm{Nklt}(X)$. We choose an (very) ample line bundle L, a smooth Hermitian metric h on L, and global sections $\sigma_1, \dots, \sigma_N$ generating $\mathcal{I} \otimes L$. Then the function

$$\rho := \sum_{\alpha=1}^{N} |\sigma_\alpha|_h^2$$

cuts out $\mathrm{Nklt}(X)$. After rescaling h, we may assume $\rho < e^{-1}$, then $\log(-\log \rho)$ is a well-defined positive function which tends to $+\infty$ near $\mathrm{Nklt}(X)$. Locally, if $\mathcal{I}_Z$ is generated by holomorphic functions $g_1, \dots, g_m$, then ρ is comparable to $\sum_{\alpha=1}^{m} |g_\alpha|^2$.

The L^∞ estimate in [DGG23, Tan26] implies that for any $\epsilon > 0$, there is a constant $C_\epsilon > 0$ such that

$$\varphi_{\mathrm{KE}} \geq -(n + \nu_X + \epsilon) \log(-\log \rho) - C_\epsilon,$$

where ν_X is an invariant depending only on X itself. In particular, we can choose a large constant $A > 0$ such that

$$\varphi_{\mathrm{KE}} \geq -A \log(-\log \rho). \tag{2.6}$$

Proof of Proposition 2.2. — The equation (2.4) implies that

$$\pi^* d\mu_h = \prod_{i \in I} |s|_{h_i}^{2a_i} \omega_W^n,$$

where $d\mu_h$ is the adapted measure associated to a smooth hermitian metric h on K_X, s_i is a global section cutting out E_i, h_i a smooth hermitian metric on $\mathcal{O}_W(E_i)$, and ω_W a Kähler form on W. Then the Monge-Ampère equation (1.1) pulls back to the following equation on W:

$$\left\langle (\pi^* \omega_X + dd^c \pi^* \varphi_{\mathrm{KE}})^n \right\rangle = e^{\pi^* \varphi_{\mathrm{KE}}} \prod_{i \in I} |s|_{h_i}^{2a_i} \omega_W^n.$$

Since the non-pluripolar Monge–Ampère measure does not charge proper analytic subsets, we have

$$\int_W e^{\pi^* \varphi_{\mathrm{KE}}} \prod_{i \in I} |s_i|_{h_i}^{2a_i} \omega_W^n = d \int_X e^{\pi^* \varphi_{\mathrm{KE}}} d\mu_X < \infty, \tag{2.7}$$

where d is the degree of f.

Then by (2.6), we have

$$\int_W \frac{\prod_{i \in I} |s_i|_{h_i}^{2a_i}}{(-\log \pi^* \rho)^A} \omega_W^n \leq \int_W e^{\pi^* \varphi_{\mathrm{KE}}} \prod_{i \in I} |s_i|_{h_i}^{2a_i} \omega_W^n < \infty \tag{2.8}$$

We now show that the finiteness of this integral will give the desired lower bound of a_i's.

Fix a point $w \in W$, let U be a small polydisk centered at w, such that $E \cap U = \{\prod_{i=1}^{p} z_i = 0\}$. Then on U we have $\prod_{i \in I} |s_i|_{h_i}^{2a_i} = \prod_{i=1}^{r} |z_i|^{2a_i} \cdot h_i^{a_i}$, where h_i's are smooth positive functions on U. We also have $\pi^* \rho = \sum_{\alpha=1}^{l} |g_\alpha|^2 h$, where g_α's are holomorphic functions that generate $\pi^* \mathcal{I}|_U$, and h is a smooth positive function on U.

We let $\mathcal{J} \subset \mathcal{O}_{W,w}$ be the ideal generated by the germs of $g_1, \dots, g_l$ at w. Then we have the following identities of germs of analytic subsets:

$$(V(\mathcal{J}), w) = \left(\pi^{-1}(\mathrm{Nklt}(X)), w\right), \qquad \left(V(z_1 \cdots z_p), w\right) = \left(\mathrm{Supp}(E), w\right).$$

By (2.5) we have $(V(\mathcal{J}), w) \subset \big(V(z_1 \cdots z_p), w\big)$. Then the analytic Nullstellensatz implies $\prod_{i=1}^p z_i \in \sqrt{\mathcal{J}}$. Consequently, after shrinking U if necessary, we can find an integer $M \geq 1$ and holomorphic functions b_α such that $\prod_{i=1}^p z_i^M = \sum_{\alpha=1}^l b_\alpha g_\alpha$ on U, then by Cauchy-Schwarz we have

$$\prod_{i=1}^p |z_i|^{2M} \leq C_0 \sum_\alpha |g_\alpha|^2,$$

for some constant $C_0 > 0$. Then (2.8) implies

$$\begin{aligned}\infty > \int_W \frac{\prod_{i\in I} |s|_{h_i}^{2a_i}}{(-\log \rho)^A} \omega_W^n &\geq \int_U \frac{\prod_{i\in I} |s|_{h_i}^{2a_i}}{(-\log \rho)^A} \omega_W^n \\ &\gtrsim \int_U \frac{\prod_{i=1}^p |z_i|^{2a_i}}{\big(\sum_{i=1}^p (-\log |z_i|^2)\big)^A} dz_1 \wedge d\bar{z}_1 \wedge \cdots \wedge dz_n \wedge d\bar{z}_n,\end{aligned}$$

here $\gtrsim$ means $\geq$ up to a constant, since we only need finiteness of the integral, such a constant does not matter. Now by Fubini's Theorem, the right hand side integral becomes

$$\int_{U\cap \mathbb{C}^p} \frac{\prod_{i=1}^p |z_i|^{2a_i}}{\prod_{i=1}^p (-\log |z_i|^2)^A} dz_1 \wedge d\bar{z}_1 \wedge \cdots \wedge dz_p \wedge d\bar{z}_p \cdot \int_{U\cap \mathbb{C}^{n-p}} dz_{p+1} \wedge d\bar{z}_{p+1} \wedge \cdots \wedge dz_n \wedge d\bar{z}_p.$$

Using polar coordinate, it follows that

$$\prod_{i=1}^p \int_0^{2\pi} \int_0^\epsilon \frac{r_i^{2a_i+1}}{(-\log r_i^2)^A} dr_i \wedge d\theta_i < \infty.$$

Then it follows easily that $a_i \geq -1$. □

Remark 2.3. — It is very likely that there is a purely algebraic argument for this claim.

Remark 2.4. — One can moreover prove that $a_i > -1$ if $\pi(E_i) \not\subset \mathrm{Nklt}(X)$. Indeed, in this case we can choose a generic point $w \in E_i$ such that $w \notin \pi^{-1}(\mathrm{Nklt}(X))$. Then $-\log \rho$ is a bounded function near w. Then the argument above implies that $|z_i|^{2a_i}$ is integrable near 0, which gives $a_i > -1$.

In summary, we get a log smooth sub-lc pair $(W, -E = \sum_{i\in I} -a_i E_i)$, and a proper surjective map $\pi : W \to X$, which is quasi-étale at x, and the Monge-Ampère equation satisfied by the Kähler-Einstein metric on X pulls back to W becomes

$$\big\langle (\pi^*\omega_X + dd^c \pi^* \varphi_{\mathrm{KE}})^n \big\rangle = e^{\pi^*\varphi_{\mathrm{KE}}} \prod_{i\in I} |s|_{h_i}^{2a_i} \omega_W^n,$$

where s_i is a defining section of E_i, h_i is a smooth hermitian metric on $\mathcal{O}_W(E_i)$. Moreover, there is an effective $\mathbb{R}$-divisor $F = \sum c_\alpha E_\alpha$ such that $\{\pi^*\omega_X\} - c_1(F)$ is a Kähler class, and $\mathrm{Supp}(F) \cup \mathrm{Supp}(E)$ is simple normal crossing.

2.3. Laplacian estimate. — We recall the following regularity result in [**BG14**, Theorem 4.6]

***Theorem 2.5* (Berman–Guenancia).** — *Let M be a compact Kähler manifold, let θ be a smooth semipositive form with $\int_M \theta^n > 0$, and let*

$$D = \sum_i a_i D_i$$

be an $\mathbb{R}$-divisor with $a_i \leq 1$. Let $F \geq 0$ be an $\mathbb{R}$-divisor such that $D_{\mathrm{red}} + F$ has simple normal crossing support and $\{\theta\} - c_1(F)$ is a Kähler class. If a full-mass θ-psh function ψ solves

$$\big\langle (\theta + dd^c\psi)^n \big\rangle = \frac{e^\psi dV}{\prod_i |s_i|^{2a_i}},$$

where dV is a smooth positive volume form, then $\theta + dd^c\psi$ is a smooth Kähler metric on

$$M \setminus (\mathrm{Supp}(D) \cup \mathrm{Supp}(F)).$$

In our setting, apply Theorem 2.5 with

$$M = W, \qquad \theta = \pi^*\omega_X, \qquad D = -E,$$

we see that $\pi^*\omega_{\mathrm{KE}}$ is a smooth Kähler metric on $W \setminus \bigcup_{i\in I} \mathrm{Supp}(E_i)$. In particular, it is Kähler on V. Consequently, ω_{KE} is an orbifold metric near x. Since x is an arbitrary point in X_{orb}, we are done.

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October 26, 2026

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